BGIN Block#11 Meeting Report

- BGIN is a multi-stakeholder discussion body / distributed think tank.

Innovation and Challenges of Blockchain Analysis

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The Global Network for Blockchain Stakeholders™

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BGIN Block#11

Innovation and Challenges of Blockchain Analysis

Date: Monday, October 22, 2024

Location: Washington DC, United States

Duration: Approx. 90 minutes

Summary

Introduction to Travel Rule Requirements: The Travel Rule in cryptocurrency introduces complexities, as digital assets like Bitcoin operate without identity requirements, making compliance challenging. The industry has developed out-of-band solutions to accommodate regulatory expectations without altering core blockchain protocols.

Industry Collaboration: In 2019, the crypto industry began working collaboratively to create a framework allowing regulated entities to share information to comply with the Travel Rule.

Challenges in Implementation: Implementing the Travel Rule is difficult due to the decentralized nature of blockchain technology, yet efforts persist to balance regulatory demands with decentralized principles.

Client Education and Risk Management: Proper configuration of risk rules and training for clients, especially those with regulatory or public sector responsibilities, are essential for interpreting blockchain analytics and handling suspicious activities effectively.

Global Regulatory Challenges: Addressing regulatory differences across jurisdictions requires a balance between local compliance and the global, borderless nature of blockchain technology. BGIN could support industry discussions by providing educational resources on these challenges.

Transparency and Competitive Concerns: Exchanges are pushing for more transparency in blockchain analytics, as current practices often lack clear evidence for high-risk scores. This “black-box” approach hinders client trust, and proprietary constraints limit open discussion.

Blockchain Analytics vs. Forensics: Distinguishing between blockchain analytics (data analysis) and forensics (evidence for prosecution) could clarify roles and improve accuracy. This distinction could benefit from an academic focus, especially regarding admissible evidence standards like the Daubert standard in U.S. law.

Emerging Proactive Detection Techniques: Proactive detection methods, including mempool analysis and opcode analysis, are evolving to preempt fraud by identifying risks before they

manifest on-chain. This anticipatory capability is unique to blockchain and not found in traditional finance.

Data Privacy Concerns: The industry's rapid advancement in analytics raises concerns about data privacy and the risk of creating a pervasive surveillance system. There is a call for more nuanced discussions around data privacy before building potentially irreversible surveillance structures.

AML and KYC in Blockchain Context: Anti-money laundering (AML) and Know Your Customer (KYC) standards in crypto compliance involve transaction monitoring and are evolving to include broader blockchain analytics and proactive fraud detection.

Philosophical and Ethical Implications: There is a debate over consumer protection in a decentralized financial environment, where some clients resist the idea of censoring transactions, believing that the responsibility lies with the users.

Staking and Block Building Responsibilities: Institutions participating in staking and block-building must consider compliance with sanctions and ethical responsibilities, such as avoiding risky or sanctioned transactions. This responsibility extends to developing compliant, competitive blocks in decentralized finance.

Emerging Market for Ethical Block Builders: As Web3 matures, there is potential for a new market of ethical block builders, akin to the sustainable investing trend. These entities could prioritize environmental and ethical standards, such as renewable energy usage and avoidance of nefarious activities.

Importance of Reliable Methodologies: BGIN could contribute by promoting accurate clustering and analytics methodologies to reduce false positives/negatives, as inaccuracies can significantly impact individuals. Case study materials may help establish a common ground for analytics providers and users.

Transcript

Green: discussion topic, Yellow: important point, Red: important^2 point

Okay, thank you. But I guess, yeah, to introduce the topic, understand the FATF requirements and of course, the description of this is specifically around the Travel Rule, we have some really interesting conversations last week at the RegTech Forum around the Travel Rule, which were fairly constructive. So I guess BGIN is a brilliant forum to continue that conversation. Of course, not everyone in the room was there. So maybe we can get started on the Travel Rule requirements so maybe pass to people here on the Travel Rule and what you think about that first. And I want this to be more like an interactive conversation.

Feel free to just put your hand up or unmute and I'll try and manage it from there.

Yeah, happy to jump in here. So the Travel Rule for anyone that's not familiar with it is essentially a requirement that was sort of inspired by the BSA rules of the 1970s, I believe, which essentially were new rules that were introduced for financial institutions to prevent the ability for illicit actors to essentially take money that had been obtained through, let's say, illicit means like the drug trade, for example, and prevent them from being able to walk into a bank and essentially launder those funds through the traditional financial ecosystem. So the Travel Rule in the traditional sense essentially just creates a data sharing between financial institutions.

Now, shortly, sort of around the time that Facebook announced the Libra project there was this sort of multilateral call for the IMF and several other organizations to look at the blockchain industry or the crypto industry specifically and figure out whether or not there was a serious degree of money laundering that could happen. And what this resulted in eventually was in 2019, the Financial Action Task Force, which is the global sort of anti-money laundering watchdog. Their goal is to minimize anti-money laundering, terrorist financing, proliferation financing globally. They created these guidelines that, and again, they have no enforcement capabilities. They're essentially an intergovernmental body that creates these standards. And it's up to the member countries to look at these guidelines and transpose them into local law.

So in 2019, we essentially took the rules from the BSA, which created this information sharing between financial institutions, and we brought that expectation to virtual asset service providers. So when it comes to blockchain analysis, that's a piece of the puzzle. The problem that you can imagine with the application of the Travel Rule is that Bitcoin, for example, doesn't care who you are. All you have to do is open a software wallet, doesn't ask for your ID or where you live, or it doesn't do any of these checks. And so what the industry needed to do to create a solution that would help us comply is we needed to create out of band mechanisms because the only options that we had were either fork your way to a solution so you can modify or attempt to modify the core rules of the blockchain. But of course, not everyone's going to agree on how to do that. And so that's not really a feasible option. And so the secondary option, the most feasible is to get the industry together to figure out if we can build something that sits on top of a blockchain. And this is what happened in 2019. The industry got together and started figuring out how we could actually create a framework that would allow companies that fall under this obligation to share information with one another.

Yeah, it's really interesting. And that was also one of my cases of the Travel Rule is that that might be difficult to implement and I think that that has been a reflection in the industry. However, there's one thing to say something is difficult to implement and one thing to not try as a result of that.

So I guess heartening to see the industry implement a fit for purpose program or schema or process for finding compliance with the Travel Rule. I guess my question around the Travel Rule

comes to do you think that in the future it can be completely blockchain enabled or like within the platforms or is it still going to require these sort of intermediary systems and service providers in order to sort of make sure that that compliance exists between VASPs.

I'm happy to jump in if that's okay.

I think you just asked the 1 billion bitcoin question. If I go back and just also clarify, the Travel Rule is something that we're not a company that provides travel solutions but many of our clients utilize our system as part of their Travel Rule apparatus. And, you know, I also want to thank you for one of the most concise and clear explanation of the Travel Rule in its history that I've ever heard. I feel like I understand it better myself now, so thank you.

But if I go back to that period when essentially the Travel Rule requirement was adopted there's a lot of frustration in the industry exactly because of the point that the moderator just raised that it's essentially in a way it's counterintuitive to the whole point. Because as you mentioned, like, you know, Bitcoin, anyone can send Bitcoin to anyone. You don't have to go through anyone.

And, you know essentially you know applying the Travel Rule would ultimately mean placing some limitations on that. And a lot of them today happen through ultimately decentralized exchanges. I think it's really important in the context of any conversation about blockchain analysis and crypto to remind ourselves that as much as we love to talk about decentralization, the vast majority of crypto trading and the way people use crypto is centralized.

Meaning that if everything was happening on a blockchain then I think to the question that it definitely would have been possible to see a day where you don't necessarily need the Travel Rule or you can do everything on the blockchain. If everything was happening in a centralized way, then it would solve the problem from the other direction, right? But then it would not be decentralized.

And the challenge here is really the fact that its usage of crypto assets, usage of decentralized assets happens across both. And, you know, that's you know, we could debate whether it will always be that way, my personal opinion is that it will forever be that way. There's never going to be a reality where all of finance or even all of crypto is just happening on chain in a completely peer to peer decentralized way and there's never going to be a time where we go back to not having it around.

So, you know, in terms of kind of, you know, the solutions, and again, it's interesting. After the effort that you describe, like when the industry started coming together, a lot of different travel solutions came out. And they're kind of like a few camps. It's interesting to see that some of those solutions are actually led by centralized exchanges that are trying to solve the problem based on their own use case, right?

Now, it's interesting to see that there are kind of a few different approaches. One of them is essentially in certain places and using certain solutions that essentially if you're a centralized exchange, you can only have your clients send funds from a wallet that's associated with you to a wallet that's associated with another exchange that is part of the same apparatus. So in some ways, you know, it's a new some would argue more efficient version of things like Swift and the fact that, you know, the reason the Travel Rule is very possible for banks is because it's a closed club, right? Like that you have to kind of go through a lot of oops, jump through a lot of hoops to be a part of. So I'm not sure if I answered the question, but the main point I'm trying to drive home is that addressing this properly in a way that both enables the decentralization and also fits within centralized regulatory framework, requires thinking about the fact that the assets will always be on both. Because both I think, are theoretically possible.

The question here is about the programmatic way to get there. So I'll stop here for a second to not take too much time and get some air, but I hope this is helpful contextualization.

Yeah, that sound really interesting. And I guess one thing that came to mind is that you almost create these bubbles of addresses within the blockchain ecosystem whichever chain you're talking about which have been tagged uh and have these sort of exchange labels next to them now. This is actually a fairly good bridge into talking to the blockchain analytics companies in the room. Do you think that these tags that are put, whether or not they're an exchange wallet or they're a nefarious wallet or whatsoever if they were available to everyone, not only the clients I guess this gets to that first question, how well equipped are clients in interpreting those blockchain analytics and the ultimate decision to flag and investigate suspicious activities lies within. I guess there's two angles to this and just bridging it from the travel conversation. One would be a flag saying that this is a travel compliant wallet that we're happy to send money to because there's a relationship between those two exchanges. And then these are also sort of the suspicious and flagged wallets that we're no longer interacting with or should be investigated and have these sorts of things.

So I'll pass to others to maybe give a bit of insight on that question of how well equipped clients of these firms are and like, I guess it's your job mostly educating on these new typologies that are coming out, there's things constantly changing.

Yeah, so just to caveat before I start, I'm not involved in the onboarding or training process myself but I am familiar with what we do in company. So whenever a client is onboarded, they have a training setup session with one of our success managers and also usually a policy or regulatory team member to ensure that they're able to configure all of their risk rules appropriately for their jurisdiction. I think that this is the most important part of client education is ensuring that those risk rules are configured correctly once those are in place then you kind of know what to do when something gets logged and you know that it's been flagged appropriately in accordance with your rich appetite and the jurisdiction that you operate.

That makes sense.

I guess, the follow-up and the agenda to that question is do blockchain analytics firms of the need to understand specific regulatory and enforcement needs of their clients, I guess, how do you go about doing that?

Yeah, I mean, that's a really good question. I mean, we certainly need to understand what our clients are looking out for. One example recently in the UK, that kind of was coming for a long time but kind of surprised us was the APP fraud rules.

So a new regulatory requirement is being leveraged against payment services firms. What it means is that a payments company, if they facilitate a transaction that ultimately leads to a fraud, as either a beneficiary or a sending counterparty, will be 50% liable for those transactions. Not a lot of crypto firms in the UK are actually regulated payment service providers, but many benefit from third parties who process those transactions on their behalf. Many clients are saying they're immediately deemed high risk by these third-party partners because they're processing transactions, onboarding, and allowing fiat to enter platforms. Yet, these providers argue there's a high risk they'll be defrauded, leading many to say they won't continue processing as third parties.

This is just one example highlighting why understanding regulatory needs and the ecosystem is crucial. This consumer protection concern is specific to the UK, and it's not necessarily a crypto regulatory issue. Blockchain analytics more broadly forms a fundamental part of the chain from identifying abuse or bad activity, which can lead to seizure, meaning that understanding how to serve clients—whether public sector investigating poor activity or private sector monitoring networks—is essential as needs evolve with regulatory pressures or internal dynamics.

We must adapt and evolve with these needs, especially in a global context where multiple geographies are concerned with ever-changing regulatory landscapes. The challenge is in balancing a jurisdiction-specific approach to solving these issues while maintaining a borderless technology model, as clients operate globally across jurisdictions.

BGIN could help by supporting the multi-stakeholder conversation and perhaps create specific educational resources, such as a document or initiative, to address the global-local regulatory dichotomy. There's a need to clarify terms, especially given the disconnect between the language and functions used by those building tools for law enforcement versus those in public sector roles.

For example, the term "scam shop" in a tool could mean different things in various jurisdictions, and these terms are sometimes defined without input from the systems using them. There's a disconnect between tools made for law enforcement by individuals not in that field, and modifying these tools can be challenging. A sunroof installed after a car is built doesn't perform

as well, and similarly, altering these tools post-creation is complex. A lack of consistency in tool terminology creates issues for financial institutions juggling multiple transaction monitoring systems. For instance, KYT systems and BSA software might be out of sync because their creators were unaware of specific regulations.

Moreover, it's tricky for those familiar with the tools to acknowledge these inconsistencies openly, as competitive concerns limit transparency. This issue extends to cases like Bitcoin Fog, which delved into whether blockchain analytics meets the evidentiary standards in court, sparking questions about data practices within blockchain analytics. While there are academic opportunities for BGIN, such as creating a non-commercial perspective on data collection practices, BGIN could help by analyzing whether blockchain analytics can replicate results and adhere to best practices.

The example of the Travel Rule showcases the challenge: a strategy designed for opaque financial instruments struggles to apply in the transparent crypto landscape. Collaborating with regulators could aid in defining data requirements, yet regulators remain largely reactive. Companies assigned high-risk scores to interactions with mixers, equating them with serious criminal activity, setting precedents that may criminalize privacy-seeking behavior. Such approaches have significant consequences for the individuals on either end of transactions.

Exchanges, increasingly frustrated with the opacity in blockchain analytics, are calling for transparency. BGIN could work with these large exchanges, who as major clients could push for changes. The exchanges' dissatisfaction with the black-box nature of analytics highlights the need for interfaces to provide evidence supporting high-risk scores, yet many companies cite proprietary reasons for withholding such evidence, despite the serious legal implications.

Creating a distinction between blockchain analytics, which is quantitative, and blockchain forensics, which involves evidence suitable for prosecution, could aid understanding. Analytics extracts information from data, whereas forensics relies on evidence for criminal cases, making it crucial to delineate what constitutes forensic data versus mere analytical data.

The Daubert standard in U.S. law, which sets criteria for admissible evidence, is relevant here, and blockchain analytics could benefit from a comprehensive analysis. The concept of blockchain analytics versus blockchain forensics and the evidentiary standards they must meet could be a focal point for an academic study, possibly as a BGIN working group topic.

Lastly, we should distinguish between evidence gathering and proactive detection, as they relate to different objectives in blockchain analytics. This proactive approach could lay the groundwork for best practices in the field, especially in collaboration with large exchanges to enhance transparency and improve user confidence.

I think evidence gathering gets much more into that forensic side. The blockchain analytics, whereas proactive detection is more about the analytics and having that ready available data. The discussion points in the agenda here is exploring two main types of blockchain analysis, which I think we've already kind of really touched on and drilled into. But I guess, yeah, are there any sort of comments from the main discussants and participants on that specific dichotomy, the evidence gathering versus proactive detection?

So let me explain that proactive detection. Yeah, so I would actually say that there's a slight difference between that dichotomy and the dichotomy that we're talking about. I see that dichotomy more as are we investigating North Korea after it happens or are we proactively going out and trying to figure out what tools criminals are using to launder their funds. I think that our company and I'm sure all the others have a mix of people focused on both sides of that. So when North Korea is wandering, when it touches the mixer, it's my job to go in and look at what mixer it is and figure out where it's changing hands and try to demix the funds. But equally, it's my job to kind of proactively go out and look at what answers are out there and what they look like and how they are being used and how they can be used.

So that's kind of slightly different in my mind to this idea of forensic investigation versus using sort of data science or other analytics methods to actually create attributions. Thank you.

Dana?

Yeah, so I can weigh in a little bit on what tools are being developed on the proactive side. The observation that I've made is most of the proactive monitoring tools are really focused on fraud. And so you've seen, we've seen an emergence of companies that are specializing, for example, in one very rich data source I think, which is underutilized is the mempool. So before the transactions are actually committed to the chain. They're hanging out in the memory pool and we've got companies now that are starting to examine what's happening in the mempool, which basically anticipates what's going to be committed to the chain.

Other companies are also doing things like op and bytecode analysis. So for example, they're able to, when a new Smart contract, for example, is constructed. They've got risk indicators to tell them what constitutes a malicious smart contract or malicious transaction. And they can look for these indicators and essentially through a variety of tools that they offer, enable you to sort of intercept those frauds before they're actually carried out. Other companies on the fraud side are doing things like monitoring for private key leaks. So when the private material is leaked, they're able to actually detect that and again intercept.

So I think that the current state of blockchain analysis, there's kind of two different things that we're looking at. One is everything that we understand about AML is basically KYC and transaction monitoring and then reporting on that. And then we have blockchain analysis with respect to looking at the chain for indicators from a broader view. What would I be interested in

from an investment standpoint? That's kind of what we're using analysis for as well. But I think the third sort of discipline that's kind of emerging pretty seriously is the fraud space. And that's where all the proactive stuff that I see is coming in.

Yeah, that's really interesting.

the pro space in the traditional finance space is fairly well defined. Is there any learning from there? We can maybe bridge into the blockchain space for these fraud tools or is that what you're describing?

So to answer the question, yes, there are learning certainly like the same kinds of frauds that have been committed for thousands of years are still being committed. It's just, you know, same story, different tool. But because we're also using a blockchain and we have such rich visibility, there's also a whole new capability that we're sort of unlocking, which is being able to look at what's happening on the chain itself, even before it's committed. So before it becomes part of the canonical history, we're looking at the indicators, what's going on on the chain, and we're able to intercept those things before they're carried out. And that is unique to blockchain. That doesn't exist in TradFi.

Is it okay to And just a couple of words.

I just want to really echo and say how inspired I am by the notion that you just iterated that, I'm going to hopefully not get too poetic but to a degree the way compliance is done in traditional finance acts like a bouncer, right? where you check out every person who tries to go into using the system. And if it doesn't pass all of the checks, they just don't go in. And then you have a group of people that you have a good idea of who they are in a room. So that reduces the risk.

And if someone does something wrong, someone also waits on the outside. And then, you know, punishes them retroactively.

And I am a firm believer that going back to your point, the transparency that the blockchain provides, it's going to take time. But it allows us to change that model. because there's combined with things like AI, combined with the opportunity to create new tools and approaches. I mean, think about it. Blockchain analytics is not an idea that existed and now there are amazing companies doing that. And, you know, like regulators talking about it and enjoying it. So there's a real opportunity to become to move towards real-time proactive ability to just to detect illicit behavior to move towards behavioral based detection and intervention rather than identity based.

Now, don't get me wrong, I don't think it's going to happen soon. I think that at the moment, for the foreseeable future, the work of a lot of crypto compliance solutions and firms is more about helping crypto firms comply with traditional compliance as much as possible. But in the long run, there's going to be more regulators who are increasingly confident about the results and how

much they can benefit from the fact that things happen on a blockchain. I think a lot of them already get it thanks to the analysis companies in the room. So just to say that I think it's fair to say that we share this with a lot of other firms innovating in this space. To me, that's the big picture. It's like, how do we create a world where people can use finance the same way they use roads?

You know, might get a little bit of training, but no one asks you before you use a road, right? There's enough confidence that if someone does something wrong, there's the right degree of monitoring so that you can intervene and get them off the road. So I hope it wasn't too poetic and high level.

But to me, that's really the kind of big goal where compliance, they would allow finance, thanks to the additional transparency and other innovation to become an open party that anyone can participate in unless you know of a kind of closed club. The key to that is more effective risk monitoring solutions, as I see it.

Not 100% related to the main topic of our discussion but How do you assess potential usefulness of blockchain analytics for more macro purposes for measuring crypto as as part of the macroeconomy, tracking for illicit activities like for example measuring or providing proxy poor close in the crypto between colonies and This is like for the one road question and one more specific just for me to understand if we look into blockchain. is there a part of activity in crypto that we are not looking at at the same moment. So if we have the data we look on addresses, on public addresses and we are then quantifying activities or flows between them considering that this is onshield and also part of this off-chain, is this a full picture of of the market that we are taking into account and if we aggregate this number what's shared like what is the fraction of the information that we get in this way is I'm asking for for even a broad answer, not necessarily very specific one, but how does it look like to show who wants to comment on that.

Yeah, we're looking for more of the macro perspective of these tools and the data sets.

I told you guys this was a topic close to my heart.

My concern from a macro perspective is that we do not have the congruent data privacy laws to be building the tools at the rate that we're building them. And I worry that as we start tokenizing everything and putting all of our interactions on chain, once you start attributing all of those interactions and risk rating all those interactions, you build a surveillance engine. And that is not a world that I want us to build. I think that we need to have a more sophisticated discussion around identity and other ways of determining credit worthiness or all of the fundamental principles that we use, like I said earlier are based off of a completely opaque instrument. It's like, how did the AML regimes as we know them today, why did they become the way that they did? Why are they as strict as they are? It's because we were dealing with a totally different beast. And it's not applicable here.

I think that we have such rich data. As I also just mentioned, we have this like anticipatory understanding of what's going to happen on chain before it actually happens. We've got the ability to connect not just what's happening on chain, but off-chain data as well. You can connect fiat data, you can connect other pieces about a person's identity through what else the wallet is doing. And yeah, in terms of like the macro view, I think we need to have some pretty serious discussions about how to preserve data privacy before we build something that we can't undo.

So I will just say the bad news is that ship sail and that's too. So if you're following some of the companies in the industry, some of the smaller startups, like there's a company out of Israel that I've been tracking that are doing exactly what you just said. or using machine learning on massive data flows to like do very high level analytics for the government agencies there's groups that are doing blockchain analytics and more flow function, doing auto tracing, doing like direct flow funds so it's not it's just go straight to what's the service in order to service and so the advances that you are talking about, I think, frankly, are already happening.I think some of them aren't as public and aren't as loud especially because they've had some time to see what happens to the companies that are really loud and really public and broadcast what they're doing with governments. And that sometimes doesn't go so well for them.

Or there's just issues surrounding that so PETA's not wrong, but what's really interesting to me was last week when we had this roundtable about dora And I'm like the door illiterate, if you will, and didn't know what it was, wasn't even on my radar And after like an hour discussion of what they were trying to do in terms of data security and operational resilience, I was like, oh.

So this is an entire regime that's trying to shove Pandora back in the box. like to your point, we are so, so behind on data privacy and data security in most countries in the world. Even GDPR was pretty late when we think about how much data was already out there. And there's still everything floating around out there and everybody's just gobbling it up.

not really caring like what they're taking, not necessarily securing what they have And if you look at Cambridge Analytics, like tip of the iceberg of the horrors of what you could do with this.

So 100% agree, but also realistic that until there's functional data privacy, until DORA goes beyond Europe, until people start to understand like the horrors of what you can do with too much data. I think it's largely going to remain small projects that go directly to governments. Like, I hope that it kind of stays that way. But I can put you in touch with some of those companies if you're interested in what that's looking like. That's like a personal project.

And this is actually related to the next session in BGIN. agenda today on building an accountable wallet ecosystem. However, using zero knowledge technology and auto protect the underlying data and the PII that would be required in order to build that accountable wallet ecosystem.

So yeah, we can continue that discussion in the next session. So really quickly, though, the proactive detection thing, the reason I asked is because I thought what that meant was When we tend to talk about blockchain analytics, or especially the way we were talking about before, it tends to be like analysis of data and we don't really talk about like where that data came from Sometimes I think the companies ingest it from the exchanges. Sometimes they get it from OSINT, I think was mentioned But sometimes there's people like running around out there like buying stuff on darknets or like interacting with criminals on telegram and kind of like throwing the machine. And if you think about how weird that is.

Like, I don't run around doing deals with drug cartels, waiting until they send me their bank number, and then posting online, hey, FYI, everyone, this bank account belongs to the Sinaloa cartel, right? Like it's a very weird dynamic. And so I was trying to picture like what you meant by proactive detection because There's proactive detection of actively going out and gathering addresses to then attribute them and see that they're illicit or gathering illicit activity and then being able to attribute that it's illicit, separated from identity, right?

Versus what sounded like we were talking about, which we talked about yesterday. Which is their white beeper about, and I'm sure there's other stuff out there, but specifically we're talking about the free chain effectively like prior to settlement using the methods that we have to detect where there's risk in the proposed transaction, so we shouldn't even allow it to get out of the mempool or be validated.

Which, yeah, I guess we would probably call it risk monitoring or pre-chain risk monitoring. I don't know that anybody's calling that. Again, is anybody calling that proactive detection?

I mean, I just think that proactive detection can apply there, but can also apply in a lot of other contexts. Only these tips.So just to give a couple of examples of ways where we oftentimes in our conversation, based on our solutions and research oftentimes we have conversations with regulators really emphasize the potential so I'll try to make it really quick. But first of all.

Rug pull scams, for example. A rug pull not something invented in crypto. The idea that you sell something and then you run away with the money, right? When it comes to scam tokens it the fact that, you know, the ultimately token is a smart contract And we've developed over time a tool called Token Sniffer that specializes in essentially scanning those smart contracts on the top 12 EVM chains. And, you know, we've basically modeled proof that you can glean within the code that basically means there's absolutely no reason that code would be there unless someone wants to run away with the money. Could be a back door.

could be you can deposit and not withdraw, et cetera. Now, as you can imagine, that leads to a huge amount of scams like we believe on average that one out of every four new tokens deployed on the chains we monitor He's a hard-coded smart contract. A hard-coded smart contract rug pull scam But you can expect that because anyone could deploy them you know you

can just copy paste the code and change a few words etc However, one thing that is really important to us to emphasize with regulators is because those assets are on-chain and are a smart contract, you can immediately detect that they're a scam the moment they are deployed before anyone has even an opportunity to buy them. So essentially, you're looking at the beginning of a reality where you have automatically detectable scams, Versus traditional finance where scams still happen, but usually the only way to do something about them is if enough people got scammed.

Another example is insider trading where the estimates of traditional finances that still like in 10, 15% of corporate news events, there's some sort of insider trading. However, when you look at insider trading in crypto and you've published some research on it, someone oftentimes what happens is that the token is available on decentralized exchanges and is listed on a centralized exchange and somewhere before the listing announcements, someone has information about it on the centralized exchange and before they announce the announcement that it's being listed on a centralized and someone's gets information about it, buys it on a decentralized exchange, it sells immediately after. You can detect that without knowing who the people are just by monitoring, you know, it's sophisticated form of monitoring that took time to develop, but just by monitoring wallet activity on DEXs compared to an off-chain piece of data, which is the listing announcement. And again, we give this as an example to regulators who obviously know what insider trading is. telling them that you see the opportunity here. this is automatically detectable insider trading. So those are to be two examples that are not necessarily in the pre-chain side of things, but to kind of give a sense.

No, that was really helpful. And I think I saw a comment that I think suggested that it's also a term in TradFi that carried over to this. It was more, I think that what I was thinking was proactive detection was different and was really directly more related to data gathering and blockchain analytics. And not, that's in the blockchain.

But not necessarily. I think it was just a terminology issue, but that's really super helpful. I think fascinating thing about what you said, and then I'll give the mic to anyone else because I'm tired of hearing myself talk but is what you're talking about from a former prosecutor perspective if we could prevent people from getting victimized by shutting it down. You can imagine a world in which we could warn a circle or a tether fast enough, right? And burn the addresses or hold the funds like as soon as before even victims could either know they were scammed or even report that they were scammed. The things that you could do here if what you're really trying to prevent is victimization or getting the money back, right?

I think the problem that I have is so much of the system when you think about criminality is tied to punishment in people, which is an identity issue that's tied to that's more closely tied to

In the sense that I've seen prosecutors ignore funds, not these funds and go after the people in non-extraditable jurisdictions. And as crypto basically becomes a place where the criminals are

very, very hard to get, but the money and the transactions may not be. I worry that like it's a massive shift in mentality to say stop caring about who committed the crime and just commit and just care that a crime was committed and just care about getting the money back for victims. And a lot of these tools are very conducive to identifying criminal activity and getting money back for victims. Regardless of who committed it, but that's like a mental shift that i think culturally is going to be really hard to focus on following the money, focus on protecting victims, focus on fighting or stopping crime.Because you're really not going to be able to stop criminals if they're sitting comfortably in Russia or China. We're never going to get that like that like that's a hard thing, I think, for people to wrap their heads around. So I think there's a fine lines that we have to think about their, especially if we're talking about blocking transactions from being confirmed, because some algorithm or some mechanism decided that that's illegal. because i'm not sure about the the term user right because there's a movie about this, right? Yeah, but not using censorship. That's a very different, horrible term. It's minority report.

right so when you de-risk somebody from a bank, we don't call it being censored out of a bank. That's the problem. Censorship is a very like that's why terminology matters. And the crypto blockchain community wants to call it because they love this Math is speech, software is speech. No, it is not. It is legally not. like you're moving money, you're moving value. That's not speech. And I think you're right. That is a fundamental like difference between if you see this as tech Yes. The people who are like, you are moving value that could be terrorist financing or criminal money or somebody's life savings. This is not an academic discussion like what you said in a newspaper or on a Facebook post. You know what I mean? Yeah, I just think it's it's uh it would be cynical for us to try and and decide or try to protect privacy. The long way we go to protect privacy But then on the other side, we prevent transactions from being executed because someone thinks could be risky. So that's the ultimate goal on privacy.

I'll just make one plug that we had this amazing conversation at DACHOM in New York about this in the white paper and Nathan Catania from Africa and I literally got into an argument about the use of the word censorship and the takeaway moral of that is you should all be at DACHOM.

So, you know, it's phenomenal conversations. But this is like, but this literally came up exactly because of the tensions between like the two mindsets and how do you balance that essentially but i'll let smarter people who have raised their hands doge

I'll make this brief. I just want to say that the gentleman who just spoke I mean, you're not wrong. Like, listen, essentially, when we have an intermediary that is dealing with someone's money, we expect because there are rules around how those intermediaries need to operate that they are going to protect their consumers.

But I work with companies on the proactive detection side, and I can tell you that they have had pushback. I mean, they've been instituted by lots of DeFi protocols, for example, who want to have a capability to get in front of frauds before they happen.

But they've also had pushback from clients who say that we believe that it's against the ethos that this industry is creating, we're not creating TradFi. So it doesn't apply to consumer protection when you're talking about peer-to-peer finance. The consumer protection obligation is on the person that's interacting in the transaction. And so they have had pushback from some clients who say, we are not comfortable censoring transactions. And so I think it is a discussion that we have to have, but the nuance here is that this isn't TradFi. So for us to keep saying like, you know, there's an obligation because it's money, well, maybe, but like the obligation is to it's it's meant to protect people. And if we're trying to create an alternative, which I believe, I mean, I'm sure everybody in this room has a different viewpoint on this and some of us will agree, some of us will disagree, but I like to believe that we can live in a world where people have options for what kind of financial system they want to participate in. And if that's the case, this is a really beautiful opportunity because you can give the user the choice. You can say, we have this wallet and we have a capability that will allow us to, when we detect fraud, censor the transaction effectively or not complete the transaction. Do you want to turn that feature on and you can let the user say yes or no. And I think that's the distinction that we sometimes forget. We have options in the system, whereas in TradFi, we don't have those options.

Because I completely agree with what you just said. I think it's a really important point that one of the crypto industry's original sins is that we started presenting it as a dichotomous alternative to traditional finance, but that's not what it's about to me. It's about expanding the realm of ways you can engage with finance. Some people can have the option to do it in a completely peer-to-peer way, and there are going to be a lot of other examples along the spectrum, and

I don't know if we have a lot of time, but if you would like, I'm happy to do a very brief presentation on the paper. All right, so this can be a much, much longer presentation. I know we're supposed to finish at four. I do have a bit of a hard cut-off, so I'll try to make it brief. I'll be very happy to take a photo op with anyone who might be interested. Just sharing my screen, confirming you guys are seeing my screen and not my messy desktop.

All right, so I'm not going to be able to go through everything, but I think a question I want to start with is: why would we even do this? A few months ago, we published a very extensive policy paper on the challenge of risk mitigation as part of the block-building process. We partnered with some of the leading experts, members of the Solidus team, and experts like Michael Mosier and Rebecca Redick, who are also involved, and Dr. Nikolai Bertentowitz.

Essentially, but I want to start with why would we even do this. We've been thinking about risk, market manipulation, and market abuse in crypto, both on-chain and off-chain, for years. As the merge was approaching, we started getting more inquiries from our institutional clients about what it means from a risk perspective. For example, let's say you are a bank or a large financial institution in a place where the regulation allows you to hold ETH. You want to use that

Ethereum, make it work for you, and participate in staking. So you become a validator, either through a service provider or you do it yourself.

Let's say that as part of the block-building process, it ends up that in a block you validated, there's a transaction that might be from North Korea or is sanctioned in another way. There's an interesting question of whether, as a validator, you are facilitating illicit activity just by the fact that you selected and validated a block with a sanctioned transaction. This question expands to other issues, like scams, fraud, CSAM, and market manipulation.

Because we don't have a lot of time, I'll just say that our paper is relatively approachable for a 55-page paper on staking. The first section talks about staking and the different kinds of staking. We then provide what we believe is the most comprehensive analysis of existing regulatory stances regarding staking and block building, including some enforcement action. For instance, we highlight what we think of as the DOJ's first enforcement action in May, where they arrested two brothers for manipulating the Ethereum ecosystem's integrity.

The big questions that institutions should consider when participating in the staking and block-building ecosystem include: what's in your blocks? How do you know there are no sanctioned transactions? How do you build competitive blocks that end up on the chain without censoring across the entire spectrum? This is something we considered when building our solution. How do you avoid censorship while allowing people to opt in or opt out based on their risk calculations?

In terms of risky blocks, we estimate that around 7.5% of all new blocks on Ethereum could potentially be deemed risky. Risky could mean involvement with OFAC-sanctioned entities or other forms of criminal activity, like market manipulation. The industry has tried to address this by excluding anything on the SDN list, but this doesn’t cover everything. For example, Nobitex, an Iranian exchange, managed to get transactions settled on Ethereum through block proposers that didn't include sanctioned transactions.

To address these concerns, we developed a Solidus builder and relay function that allows institutional stakers to opt in and receive blocks that comply with their risk requirements, like excluding sanctioned transactions. One challenge was ensuring these blocks remain competitive in terms of rewards, and so far, they are.

We built this solution to allow people to use decentralized finance in a way where risks are calculated. I apologize for the rushed presentation. This is usually a minimum 30-minute presentation, but I hope it gave a sense of what we're working on. The paper is downloadable for free on our website. I highly recommend it.

That was fantastic. I got all sorts of questions that come to mind that I wish we could get to.

This reminds me of a conversation we had about the responsibility of stakers. I've never thought about the involvement of money laundering and recently the responsibilities around staking and delegated staking. What are the users' responsibilities in terms of avoiding involvement in illicit activities? And if something happens, who is responsible?

It's a philosophical question too.

The biggest question that comes to mind is about the emergence of a new market. Just as sustainability became a market 20 years ago, we could see ethical block builders. I would want my block builder to use renewables, but this is more nuanced. I want them to avoid nefarious activities.

Yes, this is about integrity in Web3. The CFTC recently recommended that DLT be accepted as a system of record for trading, and part of this move involves tokenization. In a world with blocks built into trading systems, market integrity and fairness will become even more critical. This could be 20 years forward, but these are the thoughts I go to sleep with every night.

There’s a comment about forensic analysis and conservative approaches. How can we create a better ground to avoid these circumstances?

Chainalysis, for instance, takes a conservative approach to attribute cryptocurrency addresses to specific services, particularly if it's going to be used in a case. The Bitcoin fog judgment deemed their clustering accurate, and more conservatism might be beneficial across providers. Perhaps BGIN can help drive this initiative. False positives and negatives can impact lives, so a reliable methodology is essential, even if it varies across providers. I wonder if it would be beneficial to both providers and users of analytics if we could provide case study materials. This would cover past cases, possibilities of false positives, and establish common ground for discussion.

Yes, that could be a good starting point.

Great. Happy to be involved.

This brings us to the close of this workshop. It was phenomenal. We covered a lot of ground. Thanks to all the discussants and participants for their contributions today.